GTM965500P

Test Report

12V Model

GlobTek Engineering

Generated: April 16, 2026

Test Summary

Project: GTM965500P | 12V Model | Total Test Items: 22

Status	OK / Resolved	For Review	Regression Req'd	In Progress	Has Issue	Invalid/ Incomplete Test	Unknown
Count	4	6	1	2	0	0	9

12V

Test Item	Status	Link
Input		
No-Load Input Power	Marked for review	View →
Input Voltage	OK / Resolved	View →
Main Output		
Turn-On Delay	Marked for review	View →
Load Regulation	Marked for review	View →
Voltage Ripple	OK / Resolved	View →
Transient Response	Marked for review	View →
Startup Overshoot	OK / Resolved	View →
Hold-Up Time	OK / Resolved	View →
Remote-Sense (SNS)	Marked for review	View →
Output Voltage Fine Adjustment	Marked for review	View →
Protections		
Over-Power Protection (OPP)	Regression Req'd	View →

Changes Required

2 changes flagged across all tests.

Voltage	Test	Change Description
12V	Over-Power Protection (OPP) · *Vision314* · *2026-03-31*	I think these values are the best way to go | COMPONENT | OLD VALUE| NEW VALUE | |--------|--------|--------| | C26 | 6.8nF | 6.8nF | | C27 | 3.3nF | 2.2nF |
12V	Output Voltage Fine Adjustment · *Vision314* · *2026-04-02*	Req of Ra8, Ra9 and Ra11, Ra12 is 45.3kOhm Ra8, Ra11 = 44.2kOhm Ra9, Ra12 = 1.1kOhm Ra14 = 18kOhm Output range: 11.4V -> 12.6V

Tests Not Yet Run

11 tests have not yet been run.

12V

Test Item	Status	Link
Protections		
Over-Voltage Protection (OVP)	In Progress	View →
Short-Corcuit Protection (SCP)	In Progress	View →
Environmental / Reliability		
Component Thermal Stress	Unknown	View →
Component Electrical Stress	Unknown	View →
Safety		
Dielectric Withstand (Hi-Pot)	Unknown	View →
Output Touch Current	Unknown	View →
Earth Leakage Current	Unknown	View →
EMC		
Conducted Emissions	Unknown	View →
Radiated Emissions	Unknown	View →
Electrostatic Discharge (ESD)	Unknown	View →
Electrical Fast Transient (EFT)	Unknown	View →

Detailed Test Results

12V

12V	Turn-On Delay	OPEN	Marked for review

Specification / Procedure

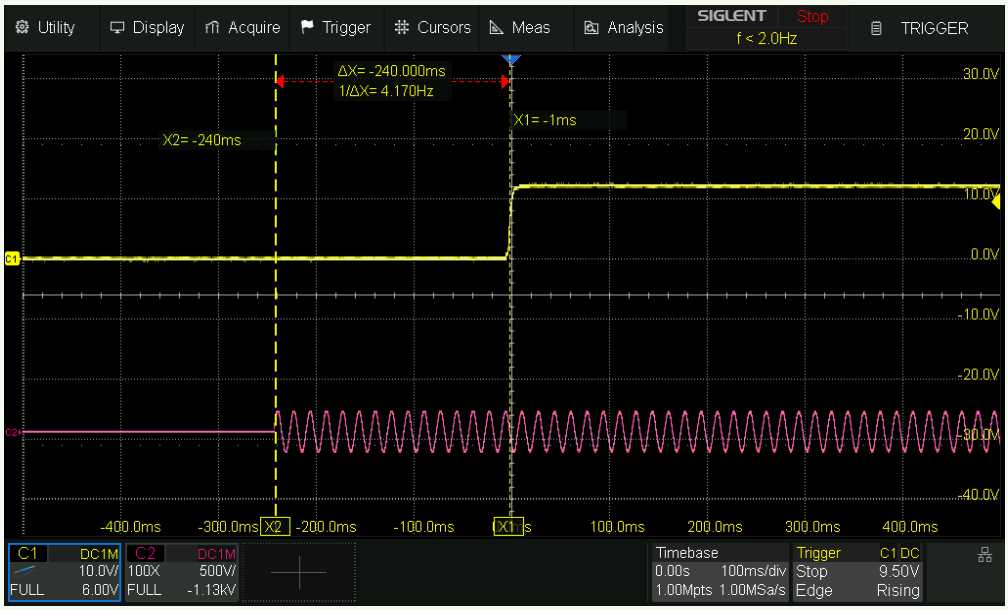
Spec: < 2 seconds @ 115Vac

Measure AC input voltage with differential probe, passive probe on DC output. Check the time between AC turn-on and DC turn-on, at full load.

Results

Vision314 · 2026-04-06

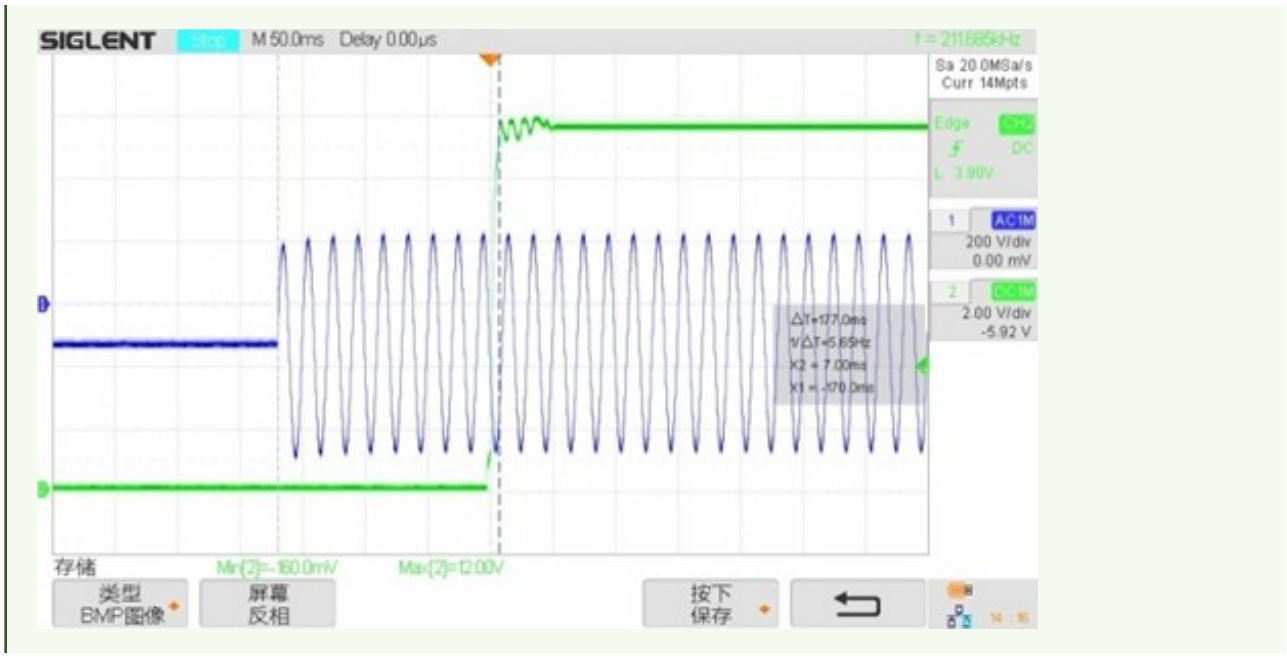
240ns @ 45.83A



Vision314 · 2026-04-16 [LATEST]

GTSZ

170ns @ 240Vac, 45.83A


SIGLENT
Stop
M 50.0ms Delay 0.00μs
f = 2116854Hz
Sa 20.0MSa/s
Curr 14Mpts
Edge
CH2
DC
L 3.90V
1
AC1M
200 V/div
0.00 mV
2
DC1M
2.00 V/div
-5.92 V
ΔT=177.0ms
1/ΔT=5.65Hz
X2 = 7.00ms
X1 = -170.0ms
存储
Min[2]=-160.0mV
Max[2]=12.00V
类型
BMP图像
屏幕
反相
按下
保存

12V	Load Regulation	OPEN	Marked for review

Specification / Procedure

Spec: ± 1.0% (measured at output connector)

Check the stabilized DC output voltage at no-load (0%). Check the stabilized DC output voltage at full load (100%).

% Load regulation = [V(no_load) - V(full_load) / V(full_load)] * 100%

Measure at 90Vac
Measure at 264Vac

Results

Vision314 · 2026-04-06

Input	V (no load)	V (full load)	Load Regulation (%)
90Vac	12.026	12.019	0.06%
264Vac	12.026	12.020	0.05%

Vision314 · 2026-04-16 [LATEST]

GTSZ

Input Voltage	12V	Load (A)	Test Result
90	12.06	0	PASS
	12	11.46	PASS
	12	22.92	PASS
	11.98	34.37	PASS
	11.98	45.83	PASS
115	12.06	0	PASS
	12	11.46	PASS
	11.99	22.92	PASS
	11.98	34.37	PASS
	11.98	45.83	PASS
230	12.06	0	PASS
	12	11.46	PASS
	12	22.92	PASS
	11.98	34.37	PASS
	11.98	45.83	PASS
264	12.06	0	PASS
	12	11.46	PASS
	12	22.92	PASS

	11.98	34.37	PASS
	11.98	45.83	PASS

12V	Voltage Ripple	CLOSED	OK / Resolved

Specification / Procedure

Spec: < 1% pk-pk (measured with a 47µF low-ESR cap + 0.1µF ceramic cap, measured @ 20MHz BW)

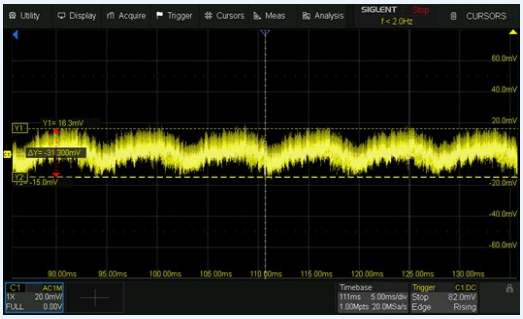
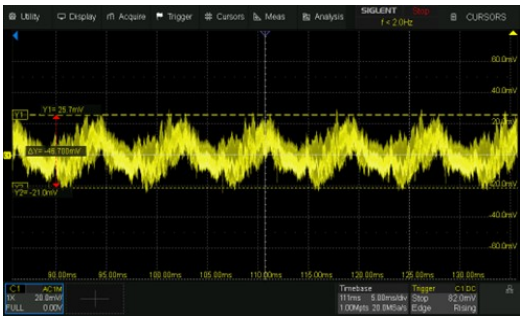
Prior to recording data, check whether 90Vac or 264Vac yields worse results, if at all. Record data using the worst case input voltage.

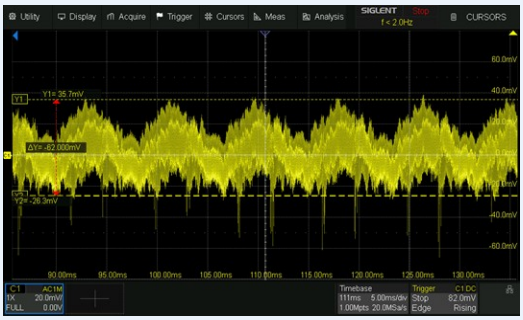
Using the ripple probe fixture and matching 1x probe, connect close to the output connectors on the PCB, measure ripple with oscilloscope. Ripple is inclusive of HF (100KHz) and LF (120Hz) components. Timebase should capture 120Hz component (e.g. 2-5ms/div)

Three conditions, adjust potentiometer accordingly:
(95% * V_out)
(100% * V_out)
(105% * V_out)

Results

Vision314 · 2026-04-06 [LATEST]
1% ripple = 120mV
ALL PASS

Output Voltage	Picture	Measured Ripple
95% Vout		31.3 mV
100% Vout		46.7 mV

105% Vout		62.2 mV

12V	Transient Response	OPEN	Marked for review

Specification / Procedure

Spec: ± 5% deviation max. (with 50% load step), recovery to within ±1% within 1ms

Taken at nominal input voltage (e.g. 115Vac)

Measure with ripple probe fixture PCB, Measure at power supply PCB. Measured with AC coupling.

Step load + load dump:
0 -> 25%
25% -> 50%
50 -> 75%
75 -> 100%

8 measurements total

Results

Vision314 · 2026-04-06 [LATEST]

Load Step

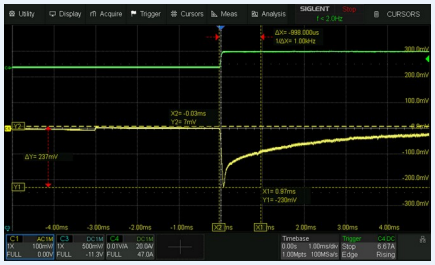
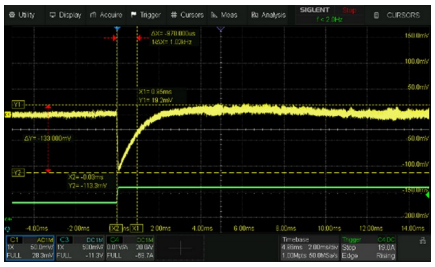
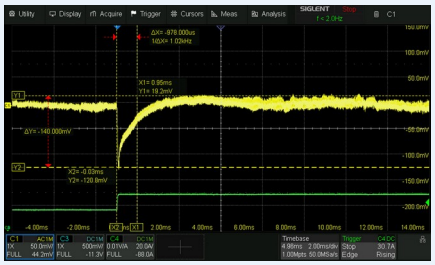
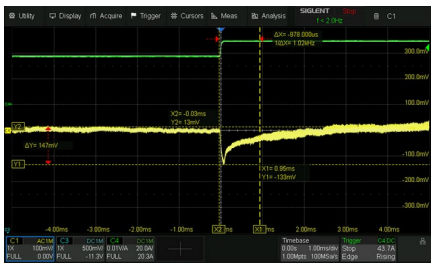
Transition	Deviation	Recovery Time
0A → 11.5A	237mV	1ms
11.5A → 22.9A	133mV	<1ms
22.9A → 34.4A	140mV	<1ms
34.4A → 45.83A	147mV	<1ms

Load Dump

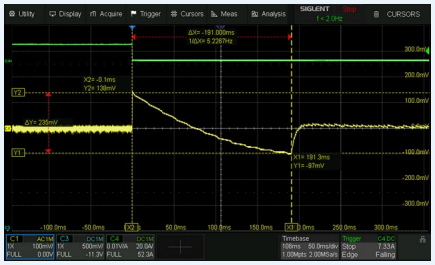
Transition	Deviation	Recovery Time
11.5A → 0A	235mV	204ms
22.9A → 11.5A	123mV	<1ms
34.4A → 22.9A	147.5mV	<1ms
45.83A → 34.4A	150mV	<1ms

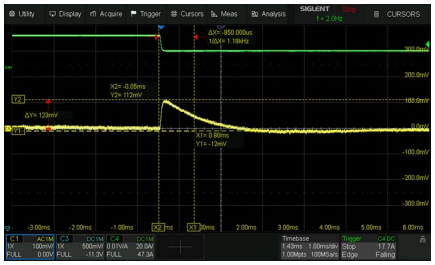
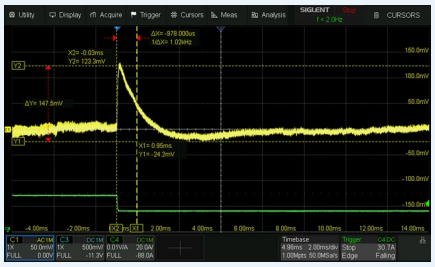
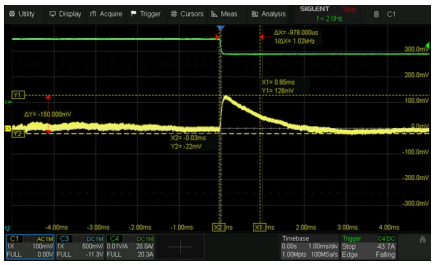
Load Step

Load Step	Picture	Deviation	Recovery Time

0A → 11.5A		237mV	1ms
11.5A → 22.9A		133mV	<1ms
22.9A → 34.4A		140mV	<1ms
34.4A → 45.83A		147mV	<1ms

Load Dump

Load Dump	Picture	Deviation	Recovery Time
11.5A → 0A		235mV	204ms

22.9A → 11.5A		123mV	<1ms
34.4A → 22.9A		147.5mV	<1ms
45.83A → 34.4A		150mV	<1ms

Comments (1)

lauxtim · 2026-04-14 [LATEST]
@Vision314 See https://github.com/GlobTek-Engineering/GTM965500P/issues/43#issuecomment-4247254592

12V	Startup Overshoot	CLOSED	OK / Resolved

Specification / Procedure

Spec: <5%

Measure with oscilloscope in DC coupling.

Measure at 90Vac and 264Vac, at:
No load
Full load

4 measurements total

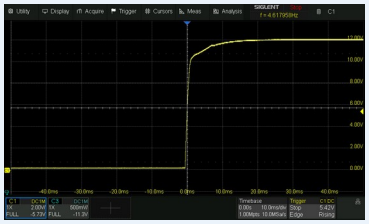
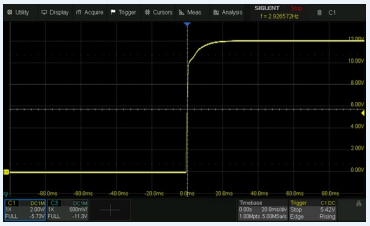
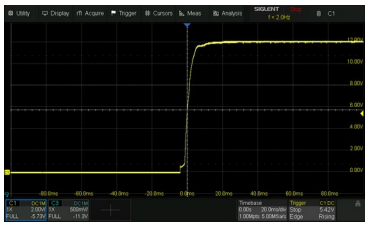
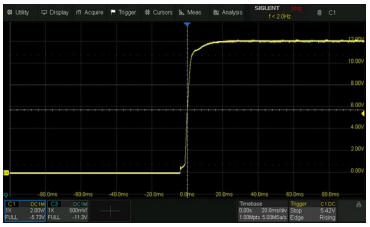
Results

Vision314 · 2026-04-06 [LATEST]

Output Voltage Overshoot

Condition	90 Vac	264 Vac
No Load	0mV	0mV
Full Load	0mV	0mV

Output Voltage Overshoot

Condition	90 Vac	264 Vac
No Load		
Full Load		

12V	Hold-Up Time	CLOSED	OK / Resolved

Specification / Procedure

Spec: 20ms typ. (100% load), 20ms min. (95% load), 50ms typ. (50% load)

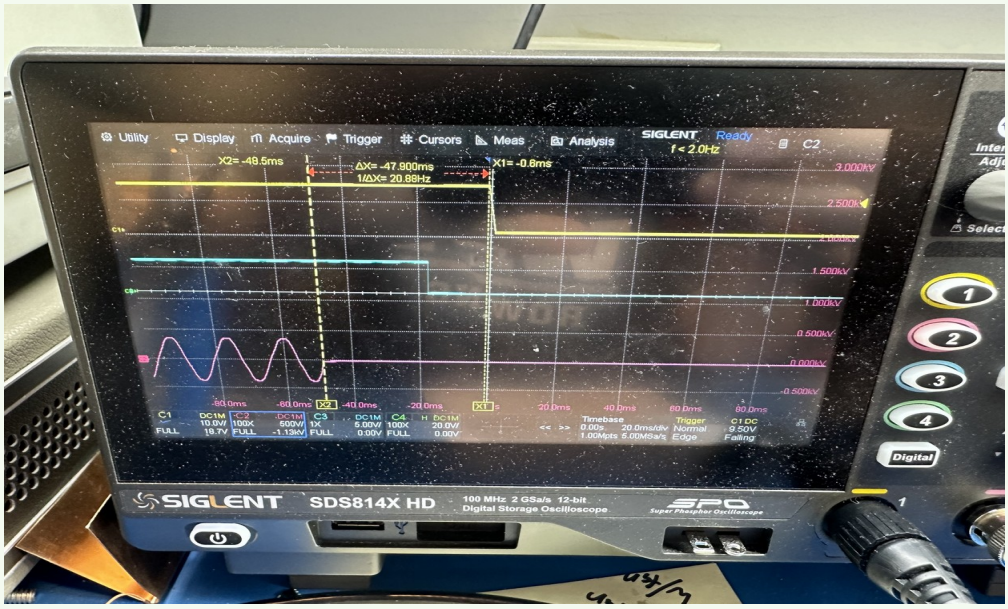
Attach differential probe to AC input. Attach passive probe to output. Set load according to the above. Measure the time between AC turn-off and DC output falling to approx. 95% * nominal V_out.

Results

Vision314 · 2026-04-02 [LATEST]

Input Voltage	Output Voltage	Hold Up Time (ms) 100% Load (45.83 A)	Hold Up Time (ms) 50% Load (22.9 A)
90Vac	11.4	25.7	54.3
	12	25.5	51.3
	13.1	23.1	46.1
264Vac	11.4	26.5	54.3
	12	25.1	51.3
	13.1	23.9	47.9

Screenshot @264Vac, 13.1V output:
Yellow - Output Voltage
Blue - PG signal
Pink - mains voltage



12V	Remote-Sense (SNS)	OPEN	Marked for review

Specification / Procedure

Spec: Up to 500mV total cable-drop compensation, reverse and short circuit protected. Connect SNS+/SNS- leads to point-of-load with twisted pair wire. (Note 1)

Attach sufficiently long wires between the output and the load, which causes a voltage drop exceeding 250mV per wire @ 85% load current. (Rationale for 85%: When remote sense is used, output voltage and output power goes up, when tested with a constant current load. We do not specify operation at P_out>100% rating.)

For instance, the DCR of 1.5m of 12AWG wire is 0.0075ohm. For the 12V model, this yields a drop of 292mV per wire, or 584mV total. Different cable lengths shall be used for other voltage models.

Part A:

- Run at nominal line voltage (e.g. 115Vac)
- Increase the set-point voltage to V_nom + 5% using the potentiometer.

1. Leave the sense leads disconnected from the load. Measure the voltage at the power supply PCB @ 85% load.

2. Connect the twisted-pair sense leads at the point-of-load (at the E-load terminals).

3. Measure the voltage at the power supply PCB again @ 85% load.

The voltage at the PCB should have increased by at least 500mV.

Part B:

1. Detach the SNS leads, reverse the leads (connect SNS+ to -load terminal and SNS- to +load terminal). Check that there is no damage during and after.

Part C:

1. Detach the SNS leads and connect them to each other (short SNS+ to SNS-, alone). Check that there is no damage during and after.

Results

Vision314 · 2026-04-07 [LATEST]
85% load = 37.1A

Part A

Step #	Voltage @ Supply	Voltage @ Load
1	12.6V	11.72 V
3	13.3 V	12.37 V

Part B - No damage

Voltage @ Supply	Voltage @ Load
13.7V	12.8V

Part C - No damage

Voltage @ Supply	Voltage @ Load
13.7 V	12.7 V

Comments (2)

lauxtim · 2026-04-07
@Vision314

> Part A
>
> 1. 12.608 V
> 2. ..
> 3. 12.86 V
>

The voltage has only increased by 252mV (if judging by 12.86-12.608). Our spec is 500mV (min.). Have you confirmed the length of wire you are using is sufficient to generate at least 500mV drop at the load current?

(You can just apply a 37.1A load (without remote sense attached) and check the difference in voltage at the PCB versus at the load to confirm.)

> * voltage @ load is only about 12.5V, should be 12.6

This might be considered "normal", as when SNS leads are connected, there is still *some* contribution from the sense points on the PCB, so we may not expect perfect regulation. Perhaps we could add a "load regulation" line to the remote sense part of the spec...maybe +/-2%.

---

lauxtim · 2026-04-15 [LATEST]
@Vision314 Good. I suggest updating the SNS results tables to something like this. This shows the "natural" drop of the cable, and then the amount of voltage the SNS circuit was able to "add back", and the spec.

Step #	Voltage @ Supply	Voltage @ Load	V_Cable drop
1	12.6V	11.72 V	0.88V
3	13.3 V	12.37 V	
V_Comp	0.70V		
V_Comp (Spec)	0.50V (min.)		

12V	Output Voltage Fine Adjustment	CLOSED	Marked for review

Specification / Procedure

Spec: Via potentiometer, ±5% adjustment range (Note 1)

11.4V -> 12.6V

Run at nominal line voltage (e.g. 115Vac)

Sense leads (SNS) disconnected.

Adjust the potentiometer to minimum and maximum positions. Record the minimum and maximum output voltage, measured at the PCB.

Changes Required

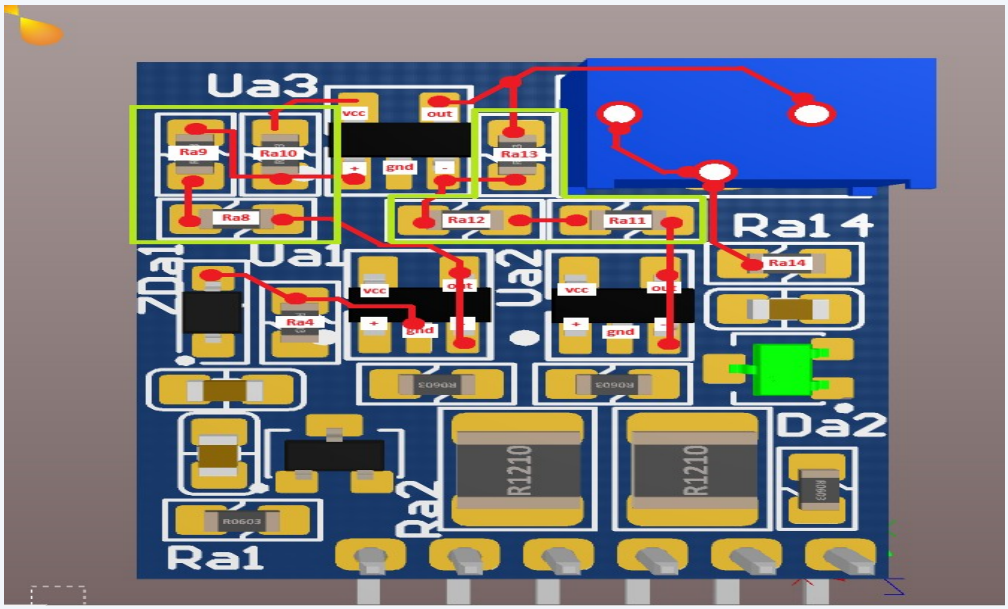
Vision314 · 2026-04-02 [LATEST]
Req of Ra8, Ra9 and Ra11, Ra12 is 45.3kOhm

Ra8, Ra11 = 44.2kOhm
Ra9, Ra12 = 1.1kOhm
Ra14 = 18kOhm

Output range: 11.4V -> 12.6V

Comments (2)

Vision314 · 2026-04-01
For future reference:



lauxtim · 2026-04-15 [LATEST]

> @Vision314 Good, can you make a separate 'result' tag for the two voltage measurements after the change?
>
> And flag the "Changes required" field as "Yes"...although I guess that might be irrelevant if we are parsing and checking for the 'change' word

12V	Over-Power Protection (OPP)	CLOSED	Regression Req'd

Specification / Procedure

Spec: 105-150%, non-latching with auto-recovery

Spec update: Change from "Over-Current Protection" to "Over-Power Protection"

Run at 90Vac and 264Vac.

The specified range applies only for V_out = V_out(nom). (e.g. "12V" @ 12.00V, "24V" @ 24.00V, etc.), voltage measured at the PCB. The nominal design point should be 130 - 135%.

Reference OPP data should also be collected at V_out(-5%) and V_out(+5% + 500mV), for reference although not specified. At V_out(+5% + 500mV), at least 100% output power should be obtainable.

Note: Initial testing was performed at 90Vac and 240Vac and at V_out = 13.3V, corresponding to V_out +5% + 700mV. This was updated in subsequent tests.

FRANKIE TESTING NOTES:

3/21/26

Noting current values for regulation board. For testing between -5% to +(5% + 500mV) => 11.4V to 13.1V

11.2 V -> 13.3V

Ra8, Ra11 = 44.2kOhm
Ra9, Ra12 = 300Ohm
Ra10, Ra13 = 10kOhm

Ra14 = 10kOhm

EDIT: make sure they are 0.1% resistors

4/2/26

OPP point will probably need another calibration after more units are produced. Seems like there is some sort of drift; when I retested it today the OPP point for max output voltage (13.1V) decrease from 47A (112%) to 46A (109%)

Changes Required

Vision314 · 2026-03-31 [LATEST]
I think these values are the best way to go

COMPONENT	OLD VALUE	NEW VALUE
C26	6.8nF	6.8nF
C27	3.3nF	2.2nF

Results

Vision314 · 2026-04-14 [LATEST]

Input Voltage	Output Voltage	OPP Point (A)	OPP Point (%)	PG Time (ms)
90Vac	11.4	68	148%	0
	12	64	140%	9.5
	13.1	47	112%	9.5
264Vac	11.4	78	170%	9.5
	12	64	140%	9.5
	13.1	47	112%	9.5

Comments (8)

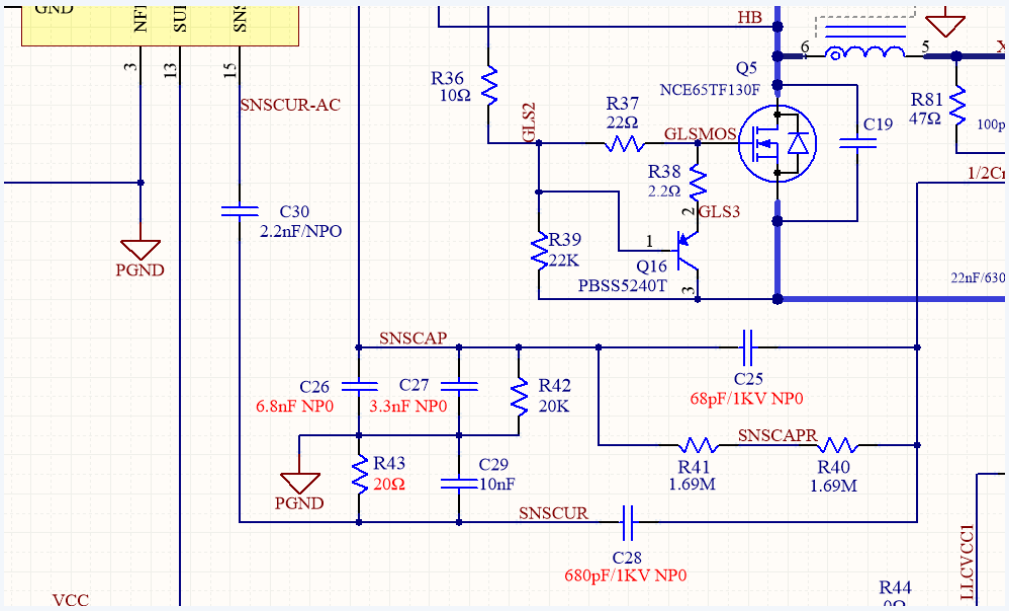
Vision314 · 2026-03-31

COMPONENT	OLD VALUE	NEW VALUE
C26	6.8nF	5.6nF
C27	3.3nF	4.7nF

OPP point is within specified range, but PG time for 90Vac typ and min output voltage is too quick. I suspect this is a problem related to the PFC at 90Vac. looking into it.

OPP Point and PG time for new values:

Input Voltage	Output Voltage	OPP Point (A)	OPP Point (%)	PG Time (ms)
90Vac	11.4	62	135%	0
	12	60	131%	0
	13.3	49	118%	10
240Vac	11.4	80	175%	10
	12	71	155%	9.6
	13.3	49	118%	9.5



lauxtim · 2026-03-31
@Vision314

> OPP point is within specified range, but PG time for 90Vac typ and min output voltage is too quick. I suspect this is a problem related to the PFC at 90Vac. looking into it.

PG time >100% load is a non-specified condition, so let's not worry about it. That said, PG time is "0" because the PFC is turning off and PFC does not signal ahead to the LLC controller about that. So, when the PFC turns off, the LLC turns off; there is no warning.

The remaining issue is that OPP @ 90Vac is different than at 240Vac, which indicates a PFC issue as noted. We could also consider this a "don't care" condition since 131 - 135% is within range, but it seems to indicate that PFC is borderline/marginally operating. We should check SNSMAINS or PFC OCP again.

Vision314 · 2026-03-31
PG time is good at 120Vac anyway.

I'll see what I can find in the PFC.

Input Voltage	Output Voltage	OPP Point (A)	OPP Point (%)	PG Time (ms)
90Vac	11.4	62	135%	0
	12	60	131%	0
	13.3	49	118%	10
120Vac	11.4	79	172%	10
	12	69	151%	10
	13.3	48	116%	10
240Vac	11.4	80	175%	10
	12	71	155%	9.6
	13.3	49	118%	9.5

Vision314 · 2026-03-31

1. Removed the temperature sensing circuit from the PFC SNSMAINS to see if it was similar to the noise issue we had on the 24V

- No change, so that's not the source of the issue
2. RINGO isn't showing any protection triggers on PFC
3. RINGO was not on the most recent version.
- Fsw min was set to 65k. Reset it to 25k (TEA2376 V9)
- no change in OPP point at 90Vac
4. Will check LLC RINGO next

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lauxtim · 2026-03-31

@Vision314

> Input Voltage	Output Voltage	OPP Point (A)	OPP Point (%)	PG Time (ms)
> 12	60	131%	0	
> 12	69	151%	10	
> 12	71	155%	9.6	

12V nominal condition at 120V or 240Vac should be set to 130-135%.

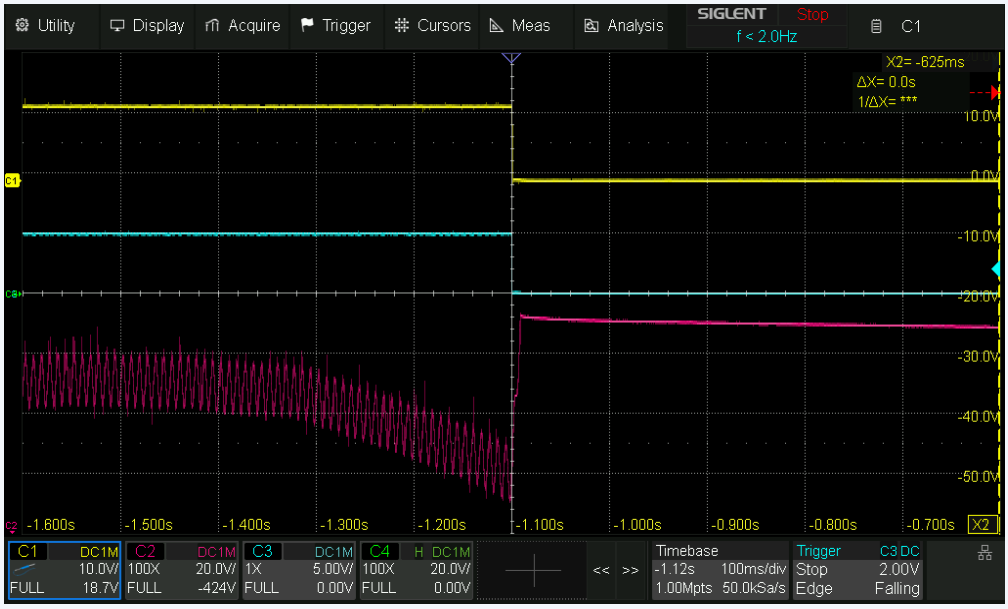
> Removed the temperature sensing circuit from the PFC SNSMAINS to see if it was similar to the noise issue we had on the 24V

Please check whether the PFC voltage starts to drop as the load approaches 130% @ 90Vac. We've seen that PFC voltage droping to ~300V causes the LLC drop out due to undervoltage and/or overcurrent due to lower PFC voltage. (Lower PFC voltage = higher LLC current.)

I would be hesitant to ascertain that SNSMAINS is not the source of the issue, even if removal of the OTP circuit does not change the situation.

Vision314 · 2026-03-31

PFC voltage is dropping before it hits OPP point at 90Vac, 12V output.
This photo is after I added the SNSMAINS rework from the 24V.



Vision314 · 2026-03-31

Replaced all the resistors in the instrumentation amplifier on the regulation board just to double check and make sure there's nothing wrong there.

> Noting current values for regulation board. For testing between -5% to +(5% + 500mV) => 11.4V to 13.1V
>
> 11.2 V -> 13.3V
>
> Ra8, Ra11 = 44.2kOhm
> Ra9, Ra12 = 300Ohm
> Ra10, Ra13 = 10kOhm
>
> EDIT: make sure they are 0.1% resistors

https://github.com/GlobTek-Engineering/GTM965500P/issues/66#issue-4147286551

Re-ran OPP test with 11.4 to 13.1 V output voltage range.

COMPONENT	OLD VALUE	NEW VALUE
C26	6.8nF	5.6nF
C27	3.3nF	4.7nF

OPP point is within specified range, but PG time for 90Vac typ and min output voltage is too quick. I suspect this is a problem related to the PFC at 90Vac. looking into it.

OPP Point and PG time for new values:

Input Voltage	Output Voltage	OPP Point (A)	OPP Point (%)	PG Time (ms)
90Vac	11.4	68	148%	0

	12	62	135%	9.5
	13.1	47	112%	9.5
264Vac	11.4	80	175%	9.5
	12	65	142%	9.5
	13.1	47	112%	9.5

---

lauxtim · 2026-04-01 [LATEST]
@Vision314

> I think these values are the best way to go
>
> COMPONENT OLD VALUE NEW VALUE
> C26 6.8nF 6.8nF
> C27 3.3nF 2.2nF
> Input Voltage Output Voltage OPP Point (A) OPP Point (%) PG Time (ms)
> 90Vac 11.4 68 148% 0
> 12 64 140% 9.5
> 13.1 47 112% 9.5
> 264Vac 11.4 78 170% 9.5
> 12 64 140% 9.5
> 13.1 47 112% 9.5

Looks good. Yes, as a general rule, if you want to give flexibility for tuning (trimming) in the future, and you have two components (either in series or parallel) which do the tuning, and both are using E-preferred values (1, 1.5, 2.2, 3.3, 4.7, 6.8), best to have one of the two be as big as possible (for coarse tuning), and the second is smaller to allow for more granular fine tuning. More granularity with the smaller values because the jump from 1 to 1.5 is small, but the jump from 4.7 to 6.8 is big, for example.

Also, shall we leave this open to indicate that we still want to take a look at the 90Vac / 11.4V PFC drop issue which has not been resolved? Or do we make a separate issue and make some sort of reference to it?

12V	No-Load Input Power	OPEN	Marked for review

Specification / Procedure

Spec: < 500mW @ 240Vac (main-output disabled, i.e. remote-off low)

Solder RO pin to AUX GND to disable output without additional regulation board load

Use Yokogawa WT3000 with 20 second acquisition time (update rate)

Results

Vision314 · 2026-04-15 [LATEST]
545 mW

Comments (1)

lauxtim · 2026-04-15 [LATEST]
See: https://github.com/GlobTek-Engineering/GTM965500P/issues/24#issuecomment-4255997812

12V	Input Voltage	CLOSED	OK / Resolved

Specification / Procedure

Spec: 90 - 264Vac (Nameplate: 100 - 240Vac)

Confirm that full rated power can be drawn over the entire rated input voltage range. Test full load @ 90Vac, 115Vac, 240Vac, 264Vac

Results

Vision314 · 2026-04-02

I think we can say this passes because of this test:

https://github.com/GlobTek-Engineering/GTM965500P/issues/66#issuecomment-4165330637

Vision314 · 2026-04-16 [LATEST]

GTSZ

Input Voltage	Output Voltage	Load (A)	Test Result
90	12.06	0	PASS
90	11.98	45.83	PASS
120	12.06	0	PASS
120	11.98	45.83	PASS
240	12.06	0	PASS
240	11.98	45.83	PASS
264	12.06	0	PASS
264	11.98	45.83	PASS

Comments (1)

lauxtim · 2026-04-02 [LATEST]

> I think we can say this passes because of this test:
>
> [#66 (comment)](https://github.com/GlobTek-Engineering/GTM965500P/issues/66#issuecomment-4165330637)

Yes, including this "Input Voltage" test is sort of a formality because many other tests will incidentally do the same test.

Untested — Specifications

12V

12V	Over-Voltage Protection (OVP)	OPEN	In Progress

12V	Short-Corcuit Protection (SCP)	OPEN	In Progress

Specification / Procedure

264Vac
Typ Output Voltage
Connect knife switch to output
When closed the supply should turn off with OCP (short circuit protection) triggered
When switch is opened again, it should turn back on.

[image: https://github.com/user-attachments/assets/d6ac253c-74a4-4190-903d-1631e99fb817]

12V	Component Thermal Stress	OPEN	Unknown

12V	Dielectric Withstand (Hi-Pot)	OPEN	Unknown

12V	Output Touch Current	OPEN	Unknown

12V	Earth Leakage Current	OPEN	Unknown

12V	Conducted Emissions	OPEN	Unknown

12V	Radiated Emissions	OPEN	Unknown

12V	Electrostatic Discharge (ESD)	OPEN	Unknown

12V	Electrical Fast Transient (EFT)	OPEN	Unknown

12V	Component Electrical Stress	OPEN	Unknown